Compare counts with the right totals

Use the denominator to see whether a changing count really means a changing rate.

Suggested teaching time: 30 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: Learners divide whole numbers and multiply by 100; formulas are supplied.

Students will be able to

- Identify the total that a count must be compared with.
- Calculate and distinguish a count change and a percentage rate.
- Write a conclusion that names the measure and stays within the figures.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; the instructor guide stays separate from student materials.
- A basic calculator is optional; learners should show the fraction and formula.
- Read all three scenarios and their model answers.
- Print the handout or display one activity at a time. Explain that all scenarios and figures are invented for reasoning practice.

Teaching sequence

5 min	**Compare two delivery records.** Learners calculate the warm-up rates. Ask why the larger late count need not mean a larger late percentage.
10 min	**Test the clinic advertisement.** Pairs calculate the count change and both appointment rates. Require the total booked visits beside each calculation.
8 min	**Separate the two claims.** Reveal the answer and use the discussion prompts. Keep count, rate, and conclusions about individual people distinct.
7 min	**Apply the method again.** Learners calculate the assignment rates independently. Review their explanation using the assessment criteria.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / More late parcels, a lower rate?

Student task and instructor answer

Explain these words as needed

Count: The number of cases, such as missed appointments.

Rate: The count divided by the relevant total; multiply by 100 to express it as a percentage.

Denominator: The number below the fraction bar: the total used as the comparison base.

Scenario

Invented scenario: A depot delivered 40 parcels on Monday, of which 4 were late. On Tuesday it delivered 160 parcels, of which 8 were late. A parcel is late if it arrives after its promised deadline.

Student output: Write two percentage calculations and one comparison sentence.

1. Calculate each day's late rate: late parcels / all delivered parcels x 100.

2. State whether the late count and late rate moved in the same direction.

Student success check: Use each day's own delivery total, not the same denominator for both days.

Model answer

Monday: 4/40 x 100 = 10%. Tuesday: 8/160 x 100 = 5%. The late count rose from 4 to 8, but the late percentage fell from 10% to 5%.

Why this meets the task: It compares each count with the number of opportunities for lateness and names both changes accurately.

Main activity / The missed-appointment claim

Student task and instructor answer

Scenario

Invented scenario: A clinic had 100 missed appointments out of 1,000 booked visits in March and 80 out of 800 in April. It advertises: 'Missed appointments fell by 20%, so patients became more reliable.'

Student output: Show the percentage change in the count and both monthly rates, then write two sentences judging the advertisement.

1. Calculate the percentage change in the missed count: (new count - old count) / old count x 100. Use (80 - 100) / 100 x 100.
2. Calculate each month's missed rate: missed / booked x 100.
3. Separate the count change from the rate change and judge whether the reliability claim follows from these figures.

Student success check: Name the denominator for each month. A lower count need not mean a lower rate.

Model answer

Count change: (80 - 100)/100 x 100 = -20%, a 20% fall. March: 100/1,000 x 100 = 10%; April: 80/800 x 100 = 10%. Fewer appointments were missed, but the missed proportion stayed the same. These figures do not support improved reliability measured by that rate, or show that any individual changed.

Why this meets the task: It verifies the true count claim while rejecting the unsupported move from a lower count to improved attendance per booked visit.

Exit ticket / More missing assignments?

Student task and instructor answer

Scenario

Invented scenario: A course recorded 18 missing assignments out of 120 assignments due in September, and 20 out of 200 due in October. The teacher says, 'More assignments were missing, so the missing-assignment rate got worse.'

Student output: Write both rates and two sentences assessing the teacher's conclusion.

1. Calculate each month's missing rate: missing assignments / assignments due x 100.
2. State which month has the higher missing count and which has the higher missing rate.
3. Explain whether the claim about a worsening rate follows, without claiming that these figures identify a cause.

Student success check: Use assignments due as the denominator; do not divide by a number of students that was never supplied.

Model answer

September: 18/120 x 100 = 15%. October: 20/200 x 100 = 10%. October has the higher missing count, but September has the higher missing rate. The claim about a worsening rate does not follow; the rate fell, and these figures alone do not explain why.

Why this meets the task: It uses the supplied totals, distinguishes count from rate, and avoids adding an unsupported causal explanation.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- Each calculation uses the relevant total for the same period.
- The percentage calculations are correct and clearly labelled.
- The conclusion distinguishes count from rate and avoids unsupported personal or causal claims.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Can the lower count still matter even when the rate stays the same?

Listen for: Yes. A count can matter for workload or capacity, while a rate answers a different question about the proportion of opportunities missed. Name the question before choosing a measure.

What would we need to say that particular patients became more reliable?

Listen for: We would need records comparing those individuals over time. Aggregate appointment totals do not identify who changed or why.

Common confusions

A smaller count always means a smaller rate. The total can also shrink. Calculate count divided by its relevant total.

The denominator can be whichever total is easiest to find. It must match the claim. For a missed-appointment rate, use booked appointments for the same period.

A falling rate shows what caused the improvement. A rate describes a pattern. Explaining its cause needs additional evidence.

Adapt the lesson

More support: Give a three-column table headed count, relevant total, rate. For March, enter 100, 1,000, and 100/1,000 x 100 = 10%. Ask learners to complete the April row using 80 and 800.

More challenge: Ask how to describe the exit-ticket rate change in percentage points. Define this as new percentage minus old percentage. Expected calculation: 10% - 15% = -5 percentage points, not a fall of 5% of the old rate.

Next practice

https://xixteen.com/learners/?practice=statistical_sense#workbench